

[EVENT]

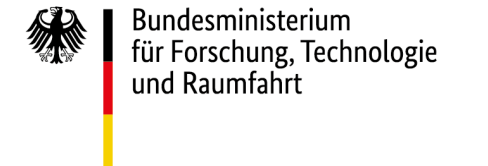
HANDS-ON RAG

Build your own document Q&A system

Felix Boelter | [Co-moderator]

AI Service Centre Berlin-Brandenburg | [Date]

Gefördert durch:



Design IT.
Create Knowledge.

www.hpi.de



PARTICIPANT LIST

The list is going round now. Please sign in while everyone is still seated.

Name	Institution	Email

The printout stays with the workshop lead. Please hand it back at the end.

START THE INSTALL NOW

It downloads about 2 GB, so let it run while we talk.

```
git clone https://github.com/
    aihpi/workshop-rag-basics.git
cd workshop-rag-basics/workshop
uv sync
```



Needs uv and Git, from the workshop-getting-started guide.

BY 1:30 YOU HAVE

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- 1 A chatbot on your laptop that answers from your own PDF.
- 2 Every claim links to the page it came from, with a score that says how well it is backed.
- 3 A list of what to check before you trust it.

AGENDA

0:00	0:15	0:28	0:41	0:54	1:10	
Welcome · setup	1 · Chat	2 · Documents	3 · Search	4 · RAG + score	5 · Your PDF · transfer	

Every notebook block: a minute of slides, then the notebook, then a quick check.

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Free workshops, talks and MOOCs

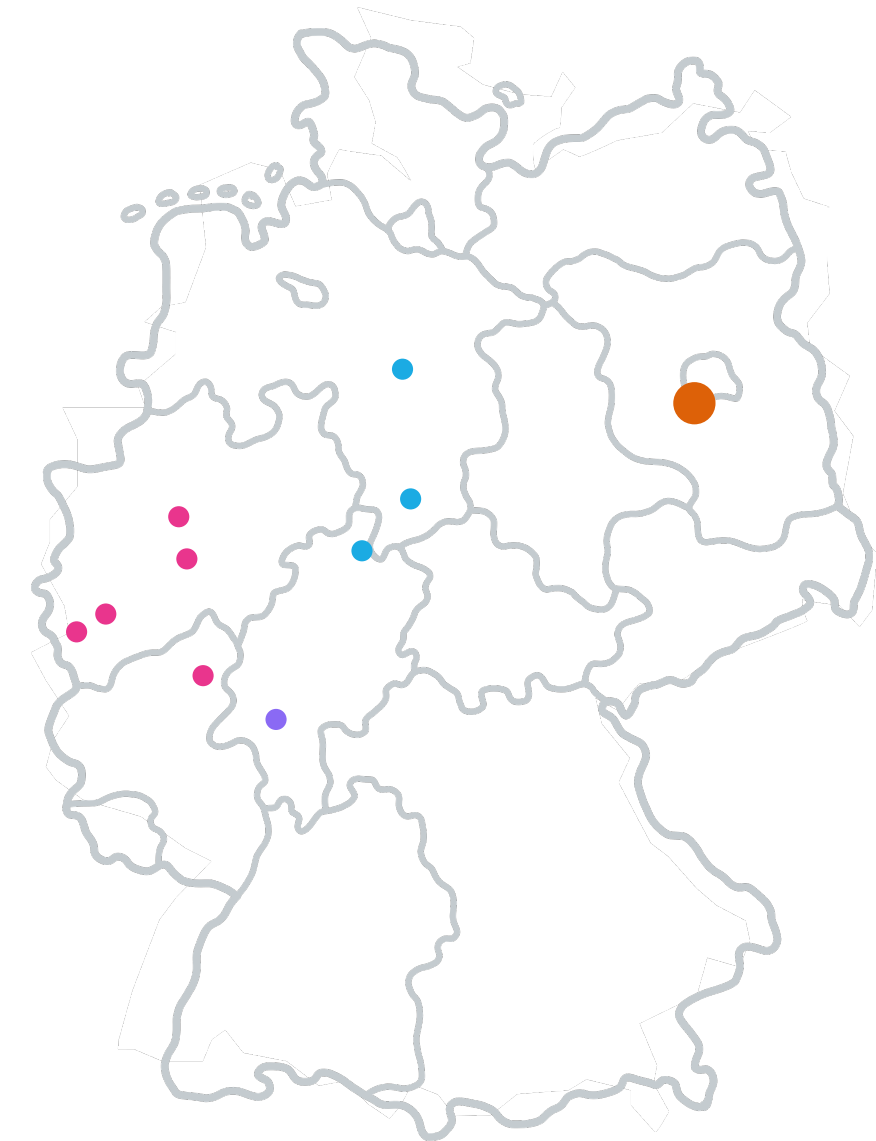
Like today, and online at openHPI

Free GPU infrastructure

For research and teaching, request access at aisc.hpi.de

Office hours and pilot projects

For your own project: more at the end



Four AI service centres, funded by the BMFTR

ARCHITECTURE

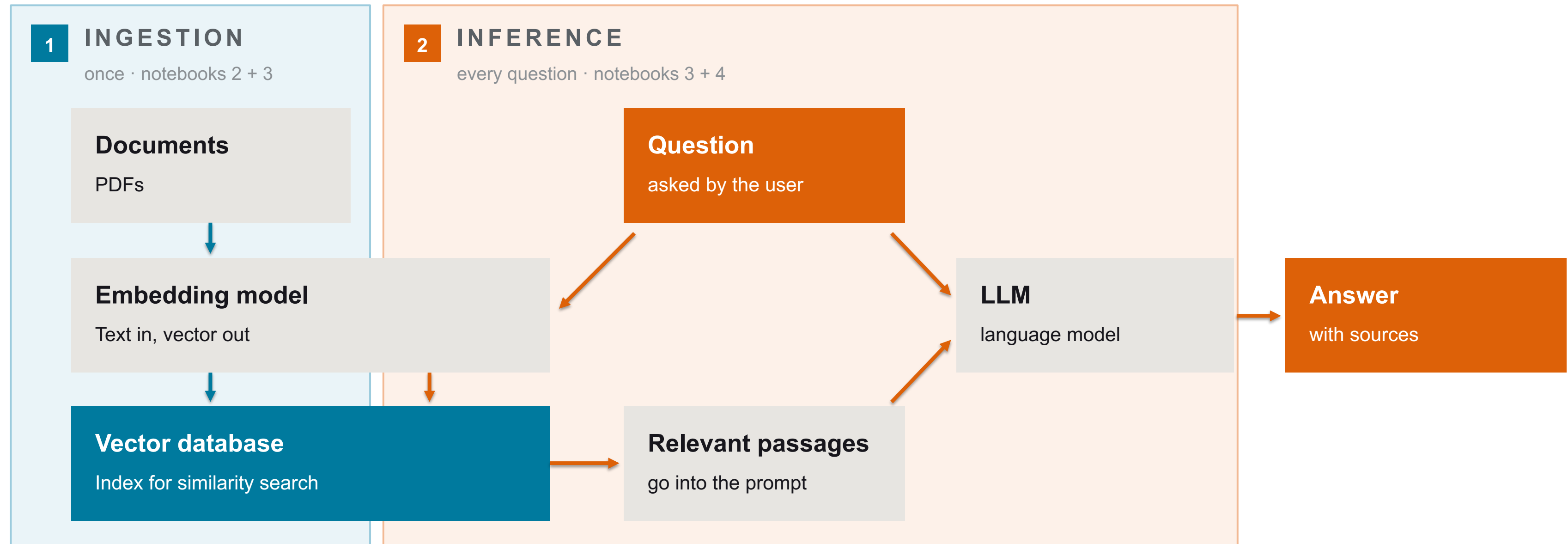
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Once, ahead of time, the documents are read and cut into chunks. The embedding model turns every chunk into a vector, a list of numbers that captures its meaning. The vectors go into the vector database.

For every question, the same model turns the question into a vector and searches the same database. The closest chunks go to the language model with the question, and it answers with sources.

SETUP

1 Install (started on slide 2)

```
git clone https://github.com/aihpi/workshop-rag-basics.git
cd workshop-rag-basics/workshop
uv sync
```

2 Add the key

```
cp .env.example .env # then paste the key from the whiteboard
```

3 Check, and fetch Docling's models

```
uv run python check_setup.py
```

4 Start

```
uv run jupyter lab
```

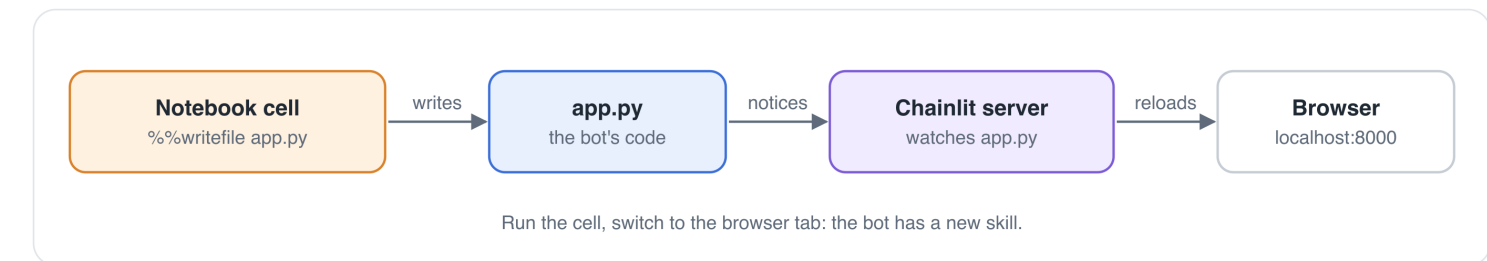
Done when check_setup.py prints **You're ready for the workshop.**

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Cells that start with `%%writefile app.py` write the chatbot. Chainlit notices the change and reloads it in your browser at `localhost:8000`.

CHAT MODEL

gemma-4-31b

EMBEDDING MODEL

octen-embedding-8b

RUNS ON

our own infrastructure in Potsdam

TALKING TO AN LLM

NOTEBOOK 1 · 6 MIN

IN THIS NOTEBOOK

- Connect to the model through the gateway
- Ask about a detail from one of our papers
- Build a chatbot that remembers and streams

YOUR TURN · IN THE CHATBOT

- 1 Personality: change the system prompt
- + Bonus: stopwatch, you'll want it in notebook 4

Solutions are folded away under each exercise.

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```
localhost:8000
```

In Kage et al. (2018), what were the lifetime-encoded beads loaded with?

✓ The model answers. Is it right?

WAS IT RIGHT?

Hands up.
Remember your letter.

NOTEBOOK 1

The model answered the Kage question. Was it ...

- A right
- B wrong
- C partly right
- D can't tell

CHECKPOINT

HALF RIGHT, AND YOU CAN'T TELL WHICH HALF

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01

No access to your documents

Language models are trained mostly on public data. Your lab reports and paywalled papers are unknown to them.

02

Similar instead of right

When the right information is missing, the model falls back on related material: the rules for no-parking zones answer a question about no-stopping zones.

03

RAG decides what the answer is built on

Retrieval-augmented generation sets which documents an answer may use, and cites the page.

READING DOCUMENTS

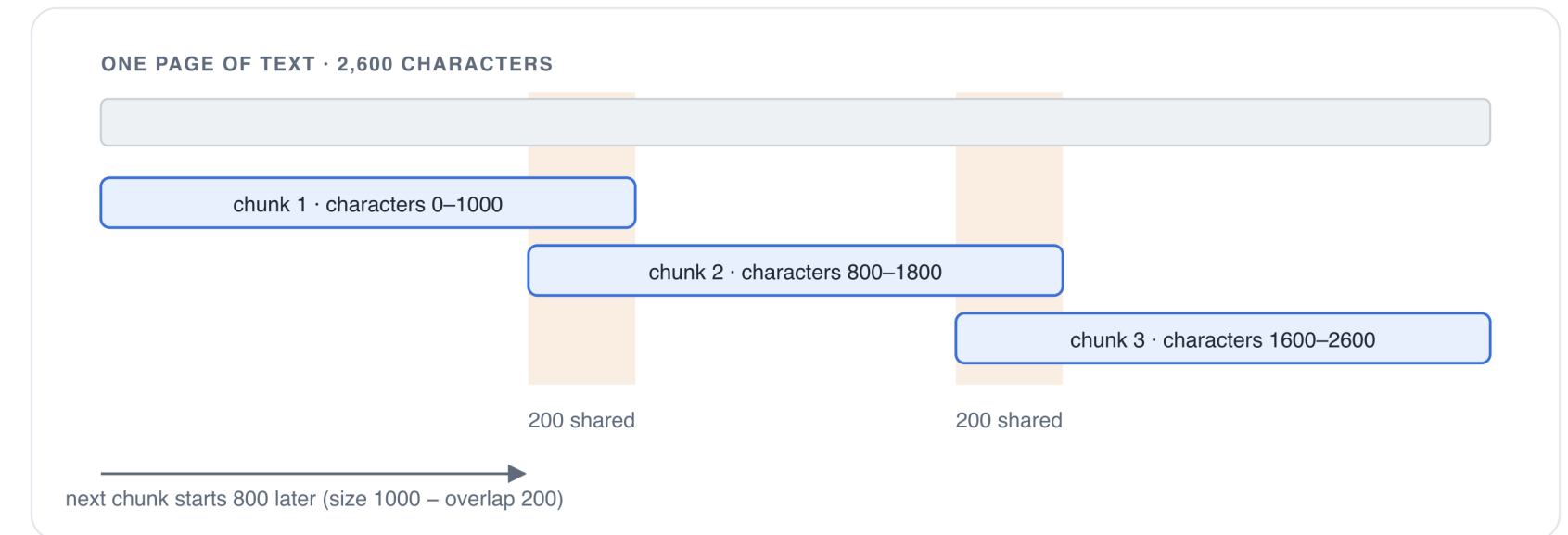
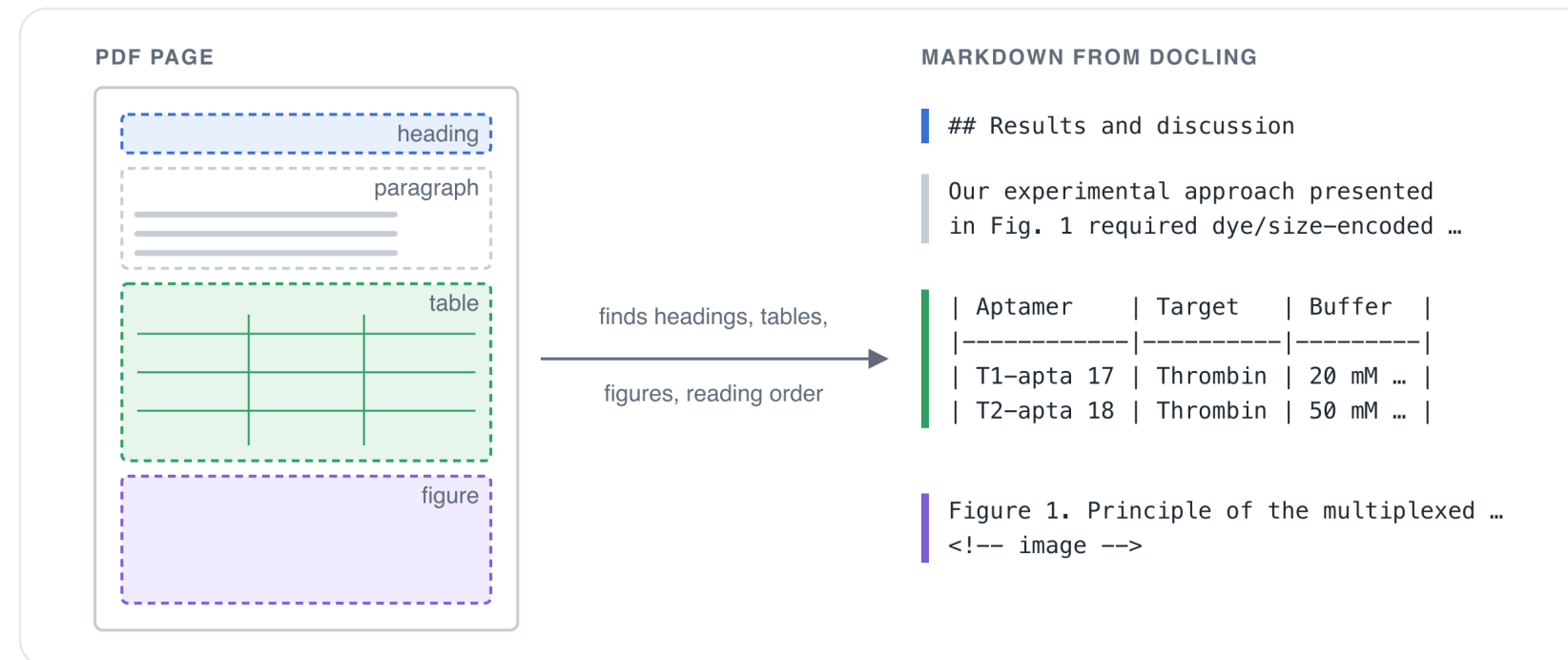
NOTEBOOK 2 · 7 MIN

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A PDF stores where to draw each letter. Docling finds headings, paragraphs and tables, and writes Markdown.

We cut chunks of 1000 characters; a new one starts every 800, so neighbours share 200.

YOUR TURN · IN THE CHATBOT

1 Where are the cuts? Try 300 and 3000

+ Bonus: only the tables

WHAT DID YOU FIND?

Vote by show of hands.
Answers on the next slide.

NOTEBOOK 2

What happened to the tables when Docling read the papers?

- A They were dropped
- B They became loose lines of numbers
- C They stayed tables, in Markdown

NOTEBOOK 2 · YOUR TURN

What happens with smaller chunks, 300 characters instead of 1000?

- A Fewer chunks, and tables stay whole
- B More chunks, and more of them cut through a table
- C Nothing changes

WHAT WE FOUND

NOTEBOOK 2

What happened to the tables when Docling read the papers?

C They stayed tables, in Markdown

All 5 tables in the three papers came through with their rows and columns, searchable like any other text.

NOTEBOOK 2 · YOUR TURN

What happens with smaller chunks, 300 characters instead of 1000?

B More chunks, and more of them cut through a table

628 instead of 197 chunks, and 17 instead of 5 end in the middle of a table row. At 3000: 69 chunks, none.

NEXT

We have good chunks. How do we find the right ones for a question?

SEARCH

NOTEBOOK 3 · 8 MIN

IN THIS NOTEBOOK

- Embed sentences and compare how similar they are
- Store all chunks in Qdrant, in memory
- Turn the bot into a search engine, no LLM yet

YOUR TURN · IN THE CHATBOT

1 Fewer, better results: pick a score cutoff

+ Bonus: ask in German

Solutions are folded away under each exercise.

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localhost:8000

What were the lifetime-encoded beads loaded with?

Kage_2018_SciReports.pdf, p. 2 (score 0.73)
› ... loaded with different organic fluorophores ...

Kage_2018_SciReports.pdf, p. 3 (score 0.64)
› Table 1. Lifetime codes and respective luminophores ...

What is the capital of France?

Kage_2018_SciReports.pdf, p. 6 (score 0.17) ...

WHY DID FRANCE GET A HIT?

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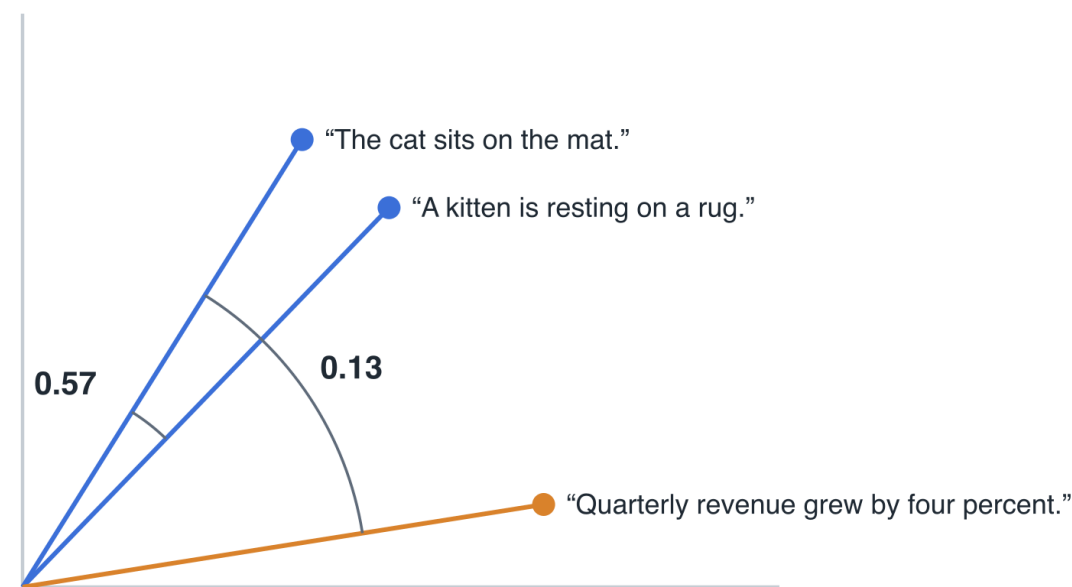


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Similar meaning → arrows point the same way.



Similarity = how closely two arrows point the same way (1 = same direction).
Real embeddings have 4096 dimensions; this drawing has 2.

Collection “papers”

id	VECTOR	PAYLOAD
0	[0.012, -0.031, 0.087, ...]	Kage_2018 · p. 1 · “Time-res...”
1	[-0.007, 0.044, 0.019, ...]	Kage_2018 · p. 1 · “multiplex...”
2	[0.021, 0.003, -0.058, ...]	Kage_2018 · p. 2 · “are loade...”

... 203 points, one per chunk

vector = what Qdrant searches by

payload = what you get back

query_points

Question → vector
embedded like the chunks

returns

5 closest points
each with score + payload

- 1 An embedding model turns text into a list of numbers, 4096 with ours. Similar meaning, similar direction.
- 2 Search returns the closest chunks, however far away they are.
- 3 Qdrant stores one point per chunk: the vector, and a payload with text, file and page.

RETRIEVAL-AUGMENTED GENERATION

NOTEBOOK 4 · 11 MIN

RETRIEVE

the 5 closest chunks

AUGMENT

the prompt with them, numbered [1] to [5]

GENERATE

the answer, citing those numbers

Every [n] leads to a chunk, and every chunk remembers its file and page.

YOUR TURN

- 1 How much context? Score limit 2 and limit 15
- + Bonus: strict or not
- + Bonus: show the evidence; follow-up questions

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WHAT THE LLM RECEIVES

system

Answer only from the context. If it isn't there, say you don't know.

user

Context:

[1] ...microbeads are loaded with different organic...
[2] ...tailor-made melamine beads with a polyelectr...
:
[5] ...imitations of lifetime encoding in flow with...

Question: what were the beads loaded with?

LLM
gemma-4-31b

The beads were loaded
with different organic
fluorophores with
lifetimes in the
nanosecond range [1].

green: the 5 chunks Qdrant found · blue: what you typed

localhost:8000

What were the beads in Kage et al. loaded with?

PMMA beads stained with organic dyes from PolyAn GmbH [1], and melamine beads loaded with CdSe/CdS/ZnS quantum dots [1], [2].

Sources: Kage_2018_SciReports, p. 2

WHICH HALF?

THE NOTEBOOK 1 ANSWER, JUDGED IN NOTEBOOK 4 (ONE RUN)

- ✓ The beads were loaded with quantum dots (QDs).
- ✗ Three different types of quantum dots, to create distinct lifetime signatures.
- ✗ One type had a short lifetime of about 10 ns.
- ✗ One type had a medium lifetime of about 25 ns.
- ✗ One type had a long lifetime of about 50 ns.
- ✗ The beads were used as fiducial markers to correct for drift during imaging.

faithfulness **0.17**

Which part of the system knows the page number of a chunk?

- A the LLM
- B the payload in Qdrant
- C the embedding model

PUTTING IT TOGETHER

NOTEBOOK 5 · 6 MIN

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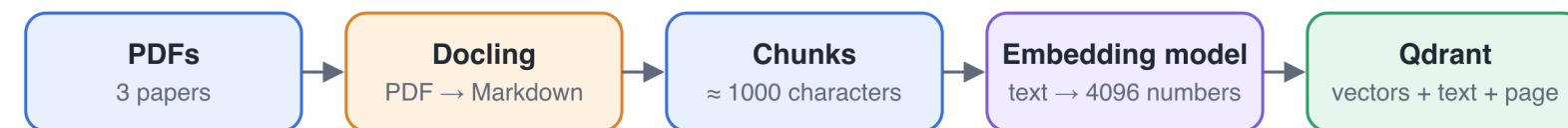
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YOUR TURN · IN THE CHATBOT

- 1 Your own document: attach a PDF, ask, check the page and the score
- + Bonus: chat with this document only

ONCE, AHEAD OF TIME



searches the same collection

FOR EVERY QUESTION



□ data □ processing step □ AI model □ vector database

PIECE · NOTEBOOK

Read a PDF with Docling	2	Cut into chunks	2	Turn text into vectors	3	Store and search in Qdrant	3
Answer from the chunks	4	Cite the page	4	Score the answer	4	Your own PDF	5

Your PDF goes to the AISC gateway in Potsdam. Test or public documents only.

YOUR COLLECTION

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QUESTION 1

Which collection would you point this at?

QUESTION 2

What do you expect to break first, and why?

One sentence each.

OFTEN

Scanned PDFs · tables · questions across several documents

WHERE TO TAKE THIS

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Today

Fixed 1000-character chunks

Qdrant in memory, rebuilt on every start

Vector search only

Faithfulness, one answer at a time

The template

Structure-aware chunkers, chosen per data source

Qdrant server, ingested once, updated when files change

Hybrid search: vectors plus keyword matching

An evaluation app that compares setups



The template

aihpi/pilotproject-rag-template on
GitHub



Office hours

Stuck on your own collection? Book a
slot.



Pilot projects

Build it with us; applications every three
months.

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YOUR FEEDBACK COUNTS

We shape our next workshops around your answers. Two minutes is enough.



kisz@hpi.de

hpi.de/kisz

Felix Boelter | [Co-moderator]

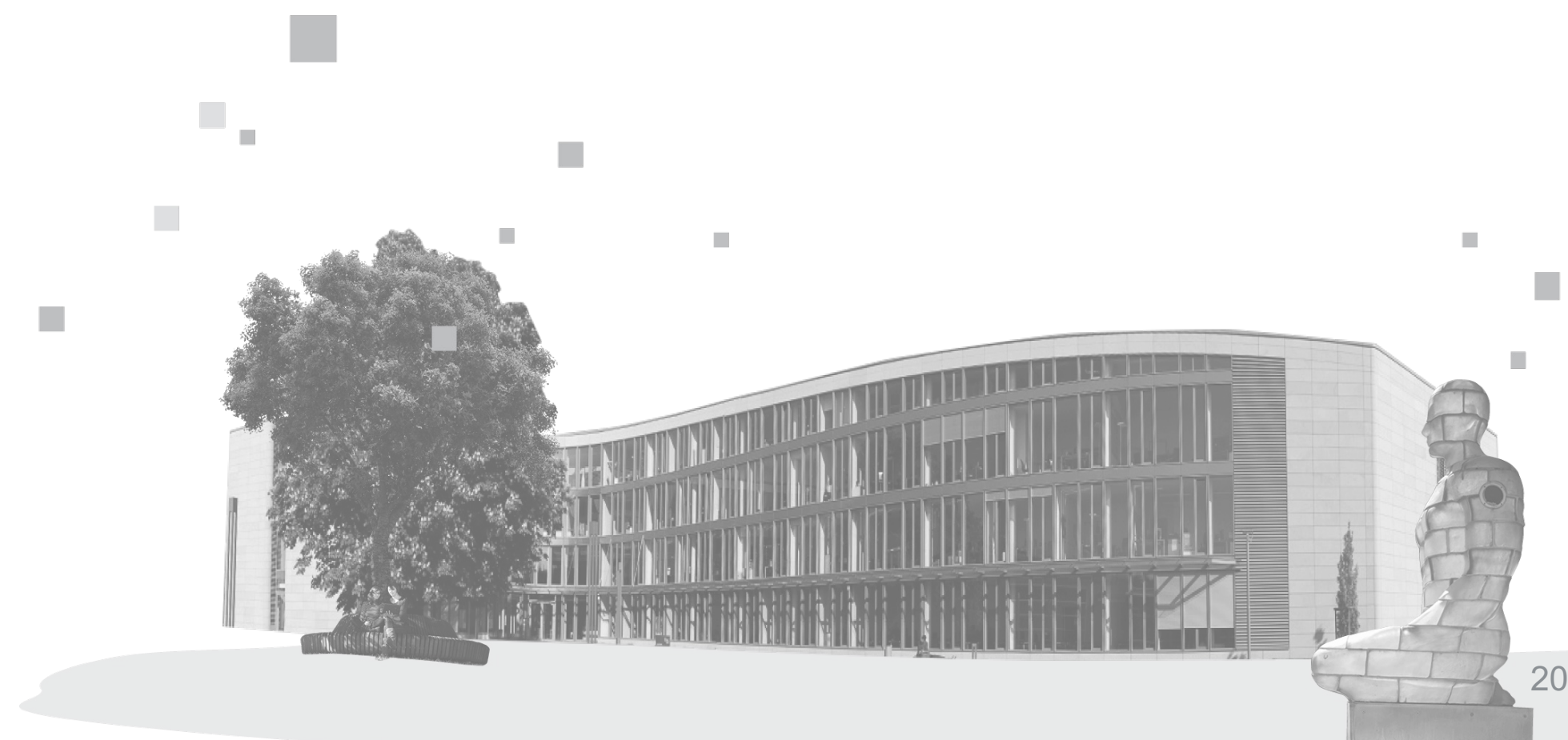
KEEP GOING

github.com/aihpi/workshop-rag-basics · this workshop

github.com/aihpi/pilotproject-rag-template · template

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APPENDIX

Background for questions. Not part of the 90 minutes.

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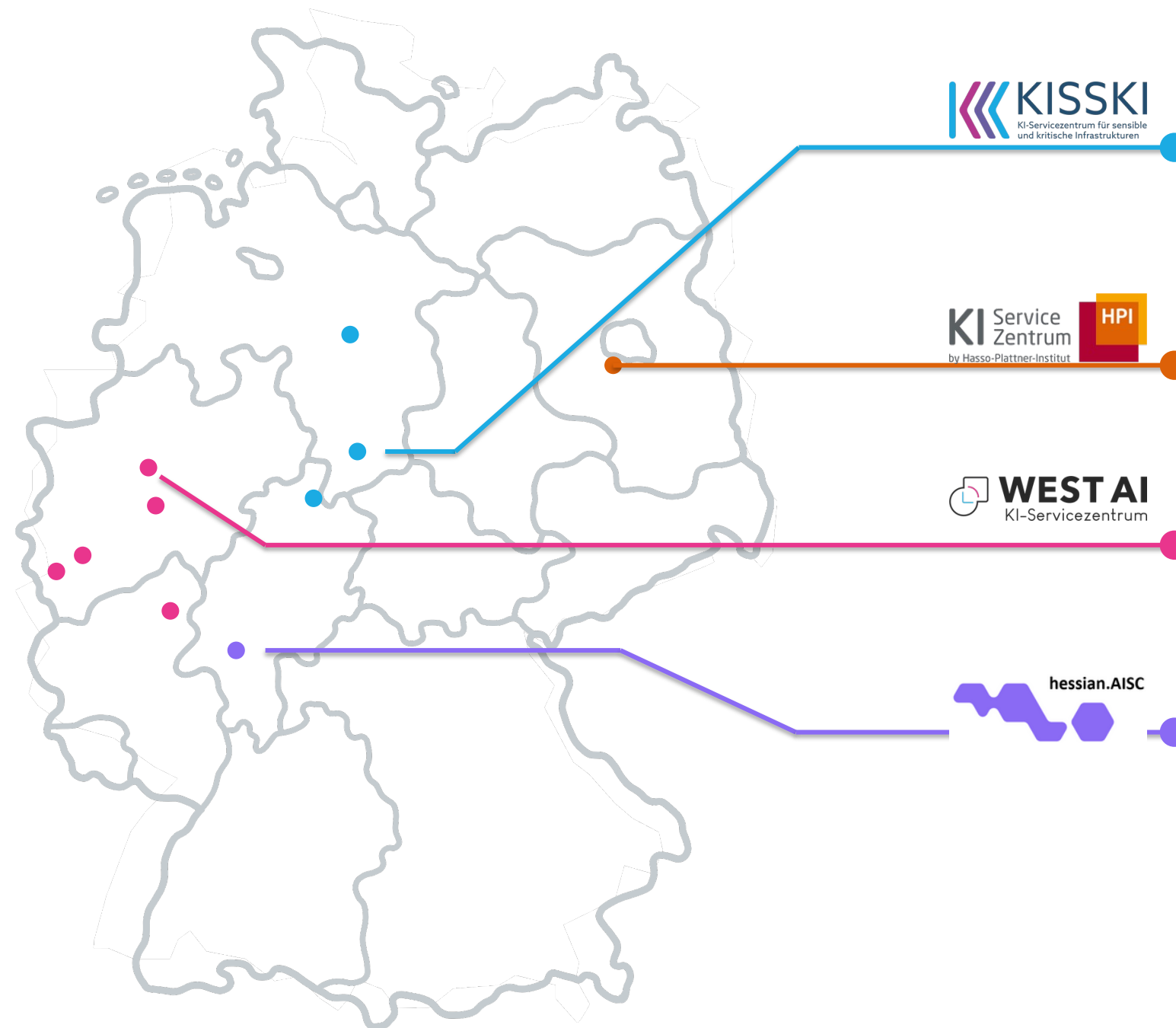


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hpi.de/ki-servicezentrum



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Potsdam | Berlin

Education and consulting, AI in business & society

WEST AI
KI-Servicezentrum

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AI hardware, consulting and training; multimodal and transferable models

hessian.AISC

Darmstadt

Explainability, generalisability and contextual adaptation



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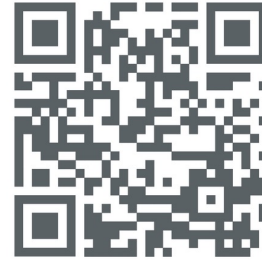
**Goal: lower the barriers to putting AI
to work in society and business**

EDUCATION



Newsletter

Talks



tele-task.de/series/1463

- Guest talks on research and innovation

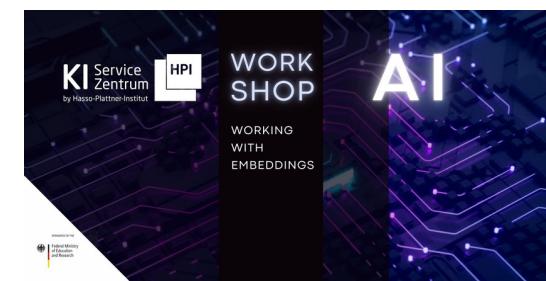


Workshops



aimaker.community

- Hands-on topics
- For example: speech2summary, Docker for ML, semantic search

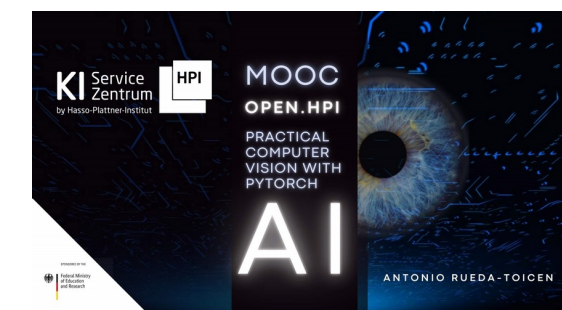


MOOCs



open.hpi.de/channels/ai-service-center

- ChatGPT: what does generative AI mean for society?
- Profitable AI
- Understanding and avoiding AI bias



INFRASTRUCTURE



aisc.hpi.de · request access



- Access **free of charge**
- No production use
 - Data should be **anonymised or synthetic**
 - No **hosting of products**
- **Reporting & publication** by users
- **Existing rights** stay with users
- **New rights** stay with users
 - Usage rights granted for research and teaching

Training

- 64 NVIDIA H100 GPUs

Inference

- 40 NVIDIA A30 GPUs

ARM server

- Ampere Altra Max M128-30 CPU
- 2× NVIDIA L40 GPUs

GPU server

- AMD Epyc CPU
- 8× NVIDIA L40S GPUs

Edge

- ARMv8 CPU
- NVIDIA Jetson AGX modules

Neuromorphic

- 288 SpiNNaker2 chips

Storage

- 1.5 PB NVRAM

Network

- 400 Gb/s InfiniBand
- 200 Gb/s Ethernet

CONSULTING



Book office hours

AI office hours

- AI infrastructure
- AI models & frameworks
- AI use cases

AI pilot projects

- Co-developing a prototype
- Applications every three months
- Selection criteria such as AI maturity and public benefit
- Results are published

Collaborations

- Jointly organised network meetings

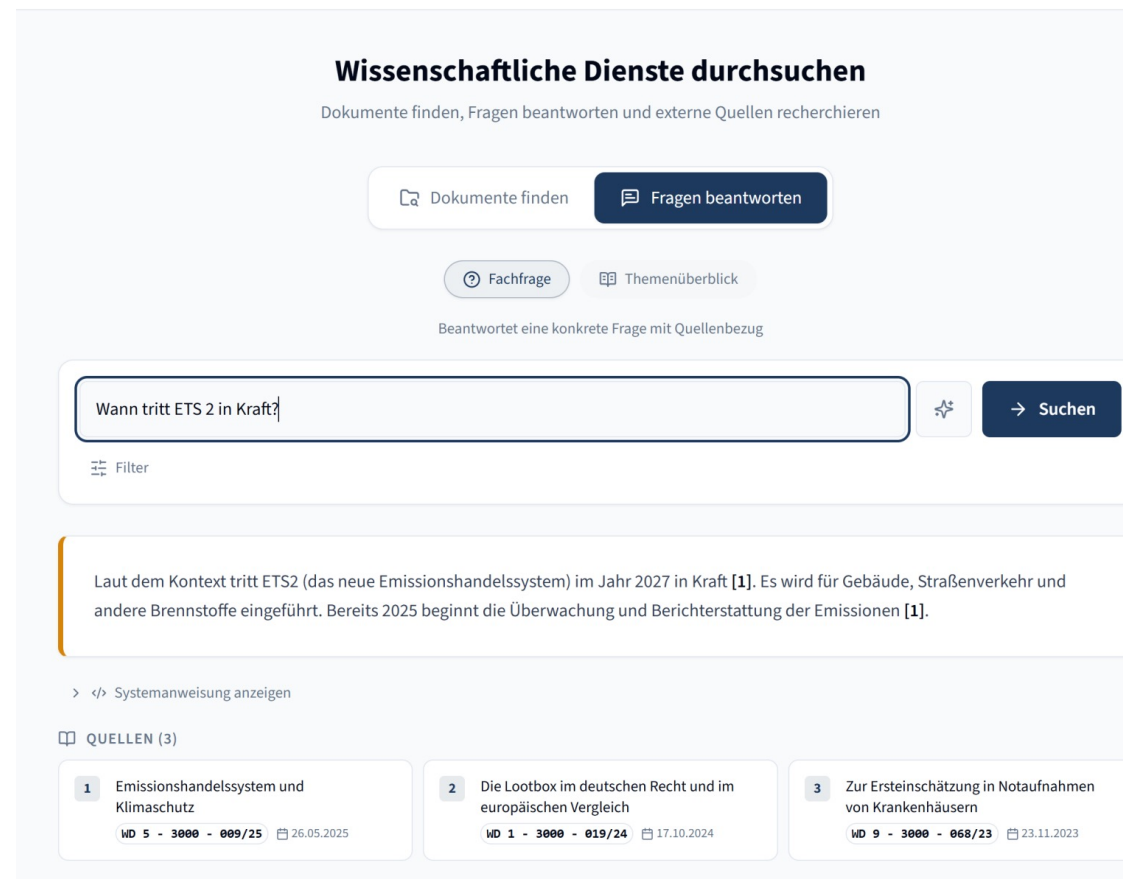
Past AI pilot projects

- Generating maths problems
- Generating plain language
- Generating upcycling ideas
- Reducing food waste
- Dating documents by handwriting



Apply now!

RAG IN PRACTICE



SENTRA

Research Services of the German Bundestag: semantic search across expert reports, an AI summary, every statement traced to its report.

github.com/aihpi/pilotproject-sentra

Semantic search · an AI summary · sources for transparency. Today you build the same thing, small.

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Was ist der Unterschied zwischen Prozess- und Systembausteinen?

Used rag_retrieve

❖ Prozess-bausteine beschreiben **Methoden, Abläufe und organisatorische Maßnahmen** des Informationssicherheits-Managements. Sie sind nach den Schichten ISMS, ORP, CON, OPS und DER gegliedert und geben jeweils Gefährdungs- sowie Anforderungsübersichten für die jeweiligen Prozesse (z. B. Risikomanagement, Notfall- und Incident-Response) (

QUELLE 1: IT-GRUNDSCHUTZ-BAUSTEINE (S.21))

System-bausteine dagegen fokussieren auf **technische Komponenten** und deren **Umgebung**. Sie ordnen sich den Schichten INF, NET, SYS, APP und IND zu und enthalten Beschreibungen von Infrastruktur-, Netzwerk-, System-, Anwendungs- und Industrie-IT Komponenten samt zugehöriger Gefährdungen und konkreter **Sicherheitsanforderungen** (QUELLE 2: DIE IT-GRUNDSCHUTZ-BAUSTEINE [S.132-133]).

Beide Bausteintypen sind modular aufgebaut, ermöglichen eine gezielte **Bausteinauswahl** und bilden zusammen das Schichten-Modell des IT-Grundschutz-Kompodiums (

QUELLE 3: BAUSTEINE (S.35)).

Bezug zu Ihren Interessen: Der Unterschied ist zentral für Ihre Arbeit an Prozess- und Systembausteinen sowie bei der Auswahl passender Bausteine im Informationsverbund.

Quellen anzeigen

Welche Kriterien nutzen Sie üblicherweise bei der Auswahl von Prozess-bausteinen für Ihr ISMS?

Wie integrieren Sie System-bausteine in die architektur Ihres Informationsverbundes?

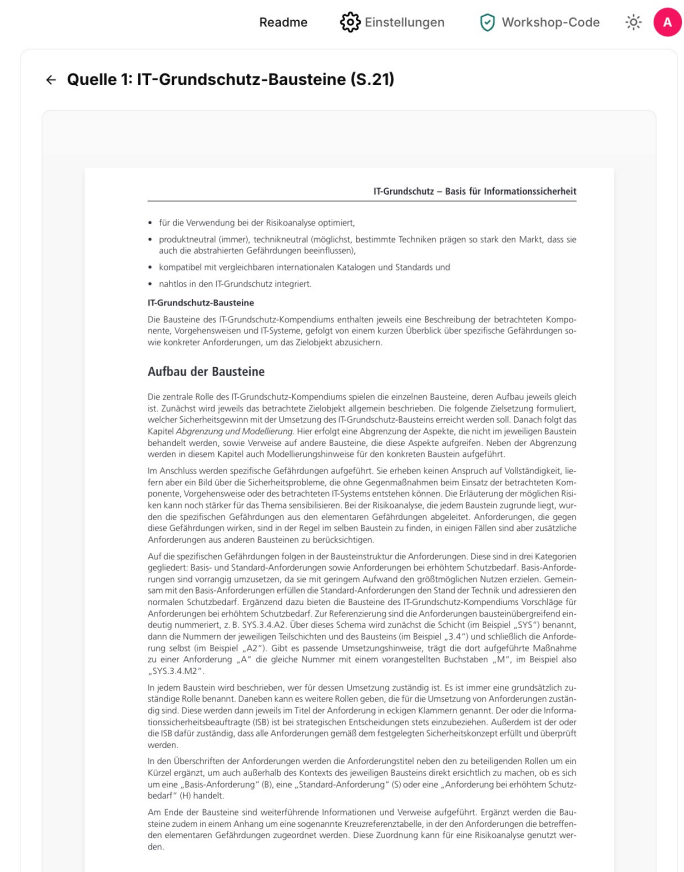
[illegible]

Type your message here..

Nutze `/export all` um alle Chats zu exportieren.



LLMs can make mistakes. Check important info



GrundschutzKI

IT baseline protection: questions in plain language, a short answer with a jump link, the BSI original open next to it.

github.com/aihpi/pilotprojekt-GrundschutzKI

FROM DOCUMENT TO CHUNKS

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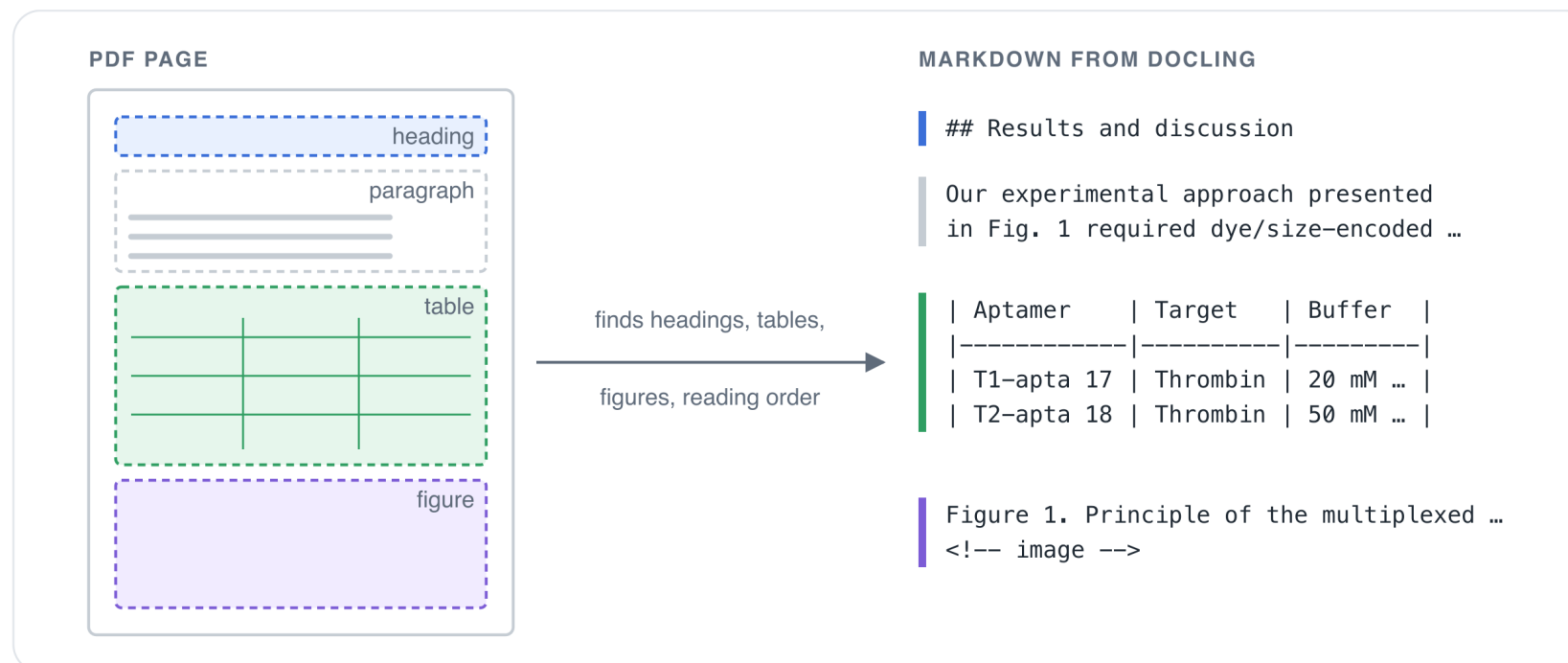


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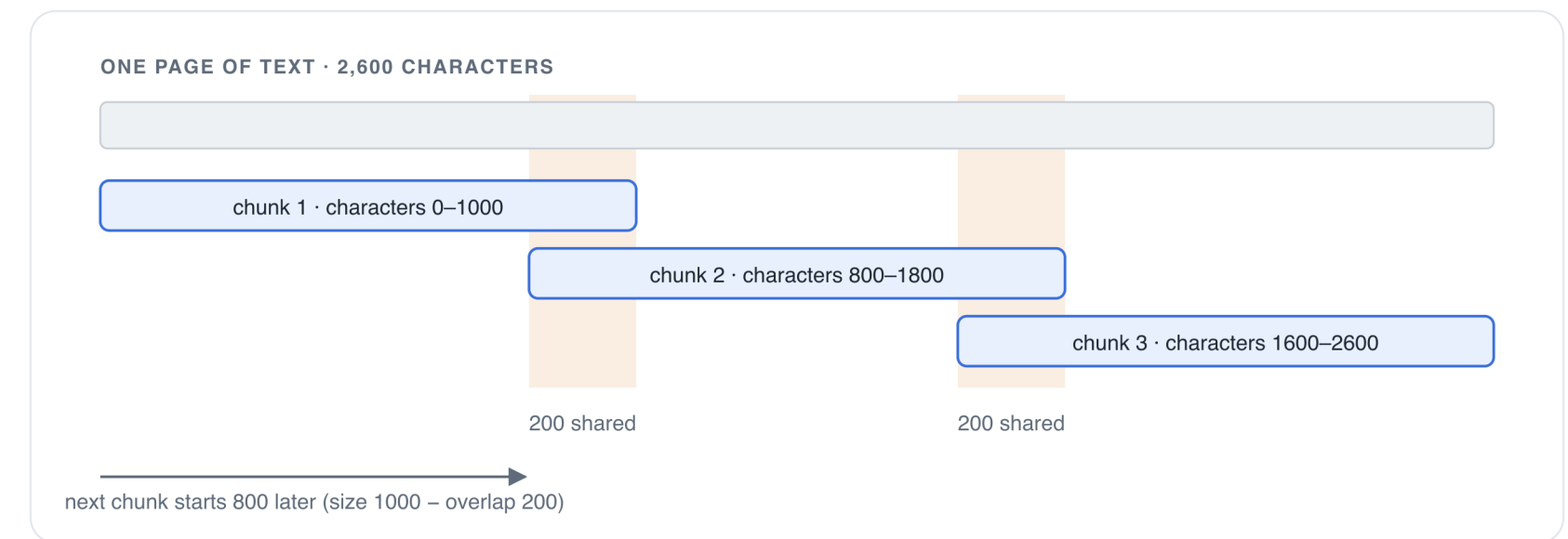


STEP 1 · DOCLING READS THE PAGE



A PDF only stores where to draw each letter. Docling finds headings, paragraphs, tables and figures, and writes Markdown.

STEP 2 · CUT INTO CHUNKS



Our notebook: every 1000 characters, with 200 overlap. Simple, the same for every document, but it may cut through a table.

GrundschutzKI: at every heading, because the BSI standards are clearly structured. A chunk keeps its title. Docling ships several such chunkers.

PROMPT AND CITATIONS

WHAT THE LLM RECEIVES

system

Answer only from the context. If it isn't there, say you don't know.

user

Context:

[1] ...microbeads are loaded with different organic...
 [2] ...tailor-made melamine beads with a polyelectr...
 ...
 [5] ...imitations of lifetime encoding in flow with...

Question: what were the beads loaded with?

LLM

gemma-4-31b

The beads were loaded with different organic fluorophores with lifetimes in the nanosecond range [1].

green: the 5 chunks Qdrant found · blue: what you typed

RETRIEVE

the 5 closest chunks

AUGMENT

the prompt with them, numbered [1] to [5]

GENERATE

the answer, citing those numbers

Every [n] leads back to a chunk, and every chunk remembers its file and page. That is all a citation is.



The number in the answer points into the list of chunks, and every chunk remembers its file and page.

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IS THE ANSWER ANY GOOD?

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You scored faithfulness in notebook 4. Changed the chunk size or the number of chunks? Measure whether it helped. Neither score needs a hand-written correct answer.

Faithfulness

Is every claim backed by the retrieved chunks?

```
supported claims ÷ all claims
```

A judge model splits the answer into claims and checks each one against the chunks. 0.5 means half the claims have no source.

Relevance

Does the answer address the question?

```
similarity(questions from the answer, real question)
```

A judge model writes the questions this answer would fit; their embeddings are compared with the question that was asked.

In the template: both scores for every answer, a badge in the chat, and an evaluation app that compares configurations. Built on Ragas.